# Heart Disease Prediction: End-to-End MLOps

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**Course:** MLOps (AIMLCZG523), Assignment 01
**Repository:** https://github.com/sreejarepo/MLops **Date:** 18-10-2026
**Demo video:** https://drive.google.com/file/d/19tLMqSR8QTFi2KamsZ3QJQGIrb-G8ACh/view?usp=sharing

## 1. Abstract and objectives

This project builds a binary classifier that predicts the presence of heart disease from the 13 clinical features of the UCI Heart Disease dataset, and delivers it as a monitored, containerised API. The objectives were to make every step reproducible and automated: data acquisition, cleaning, EDA, training with experiment tracking (MLflow), tests and CI (GitHub Actions), packaging (Docker), deployment on Kubernetes (Minikube) and monitoring with Prometheus and Grafana.

The deployed model is a tuned Random Forest with a 5-fold cross-validated ROC-AUC of 0.905. It is served by a FastAPI service whose `/predict` endpoint returns the predicted class, a label and a confidence value.

## 2. Dataset and acquisition

The data is the UCI Heart Disease dataset, **ID 45** (https://archive.ics.uci.edu/dataset/45/heart+disease), downloaded programmatically by `download_data.py` through the `ucimlrepo` package, with no manual download. The script retries three times and falls back to the committed CSV if the UCI server cannot be reached. The result is **303 rows x 13 features** plus a target column. The original target (0 to 4) is binarised: 0 means no disease and 1 to 4 means disease present, which gives **164 patients without disease and 139 with disease**. The data comes from the UCI Machine Learning Repository, and the original donors include the Cleveland Clinic Foundation.

## 3. Data quality and EDA

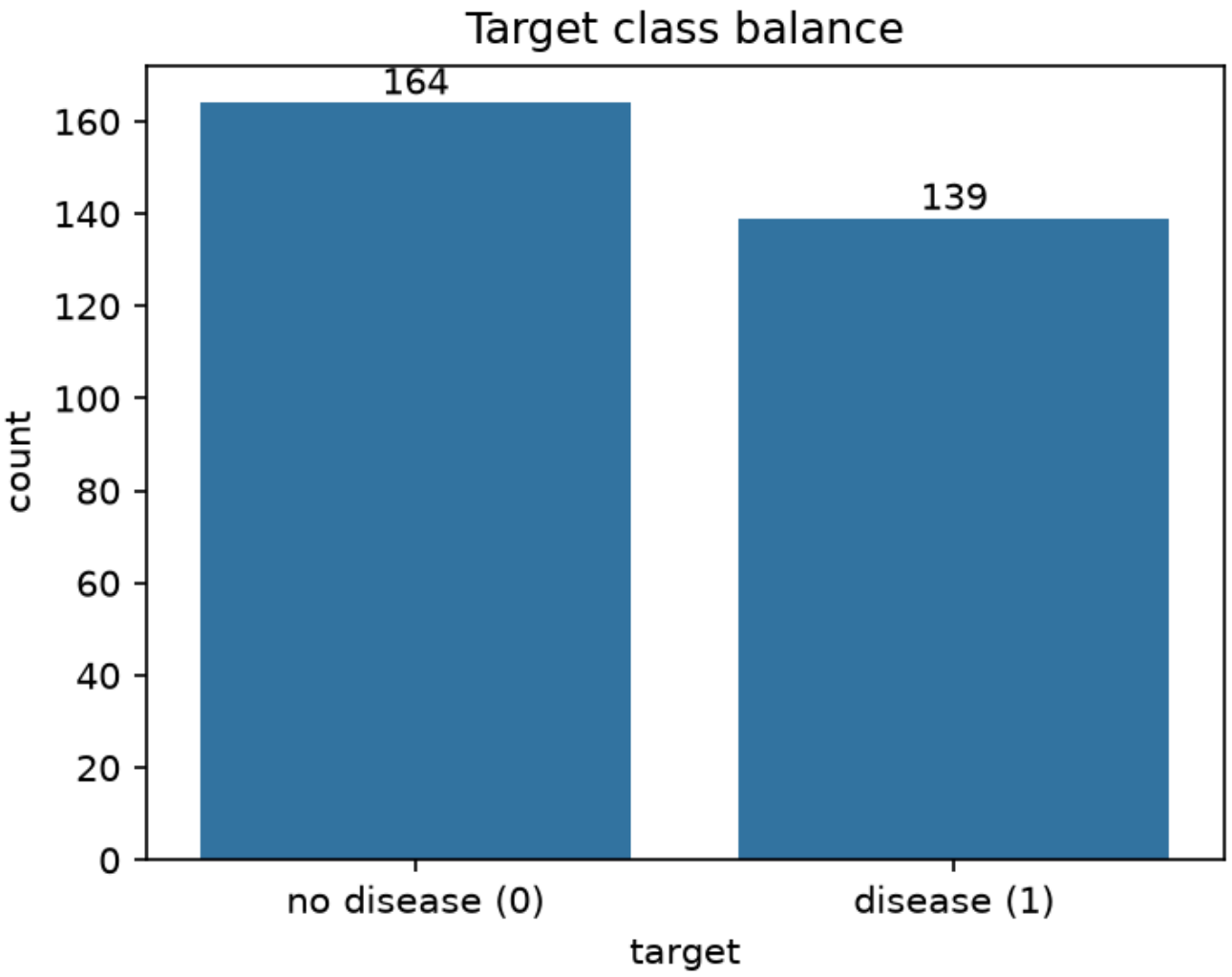
`src/clean.py` writes a cleaned copy to `data/processed/` and a `data_quality_report.json`. The quality checks found:

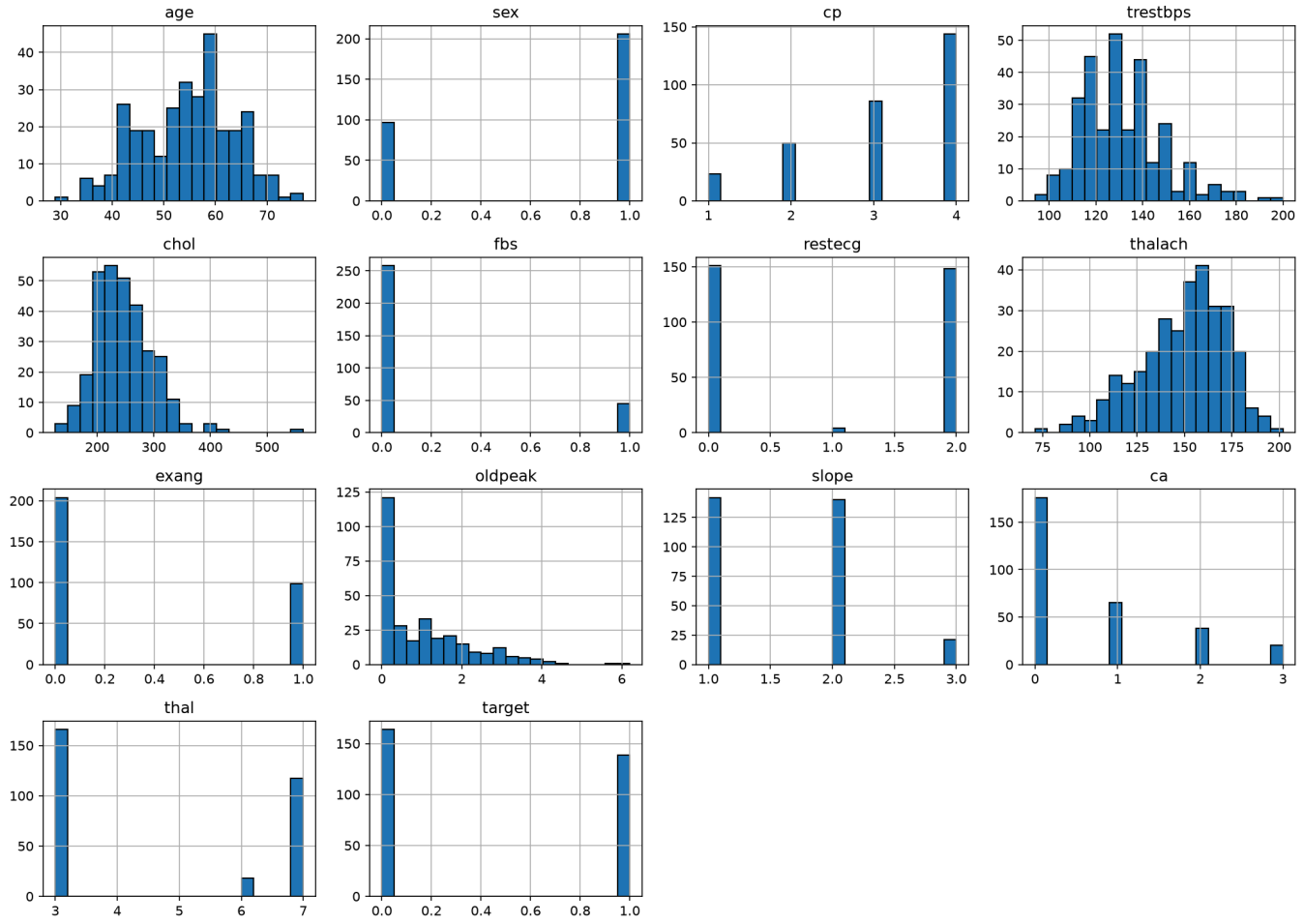
- **Missing values:** only two columns, `ca` (4 rows) and `thal` (2 rows). They are imputed with the most frequent value (`ca` = 0, `thal` = 3), after which there are no missing values.
- **Duplicates:** none, so all 303 rows are kept.
- **Class balance:** 164 vs 139 (54.1% vs 45.9%), balanced enough that I do not resample.

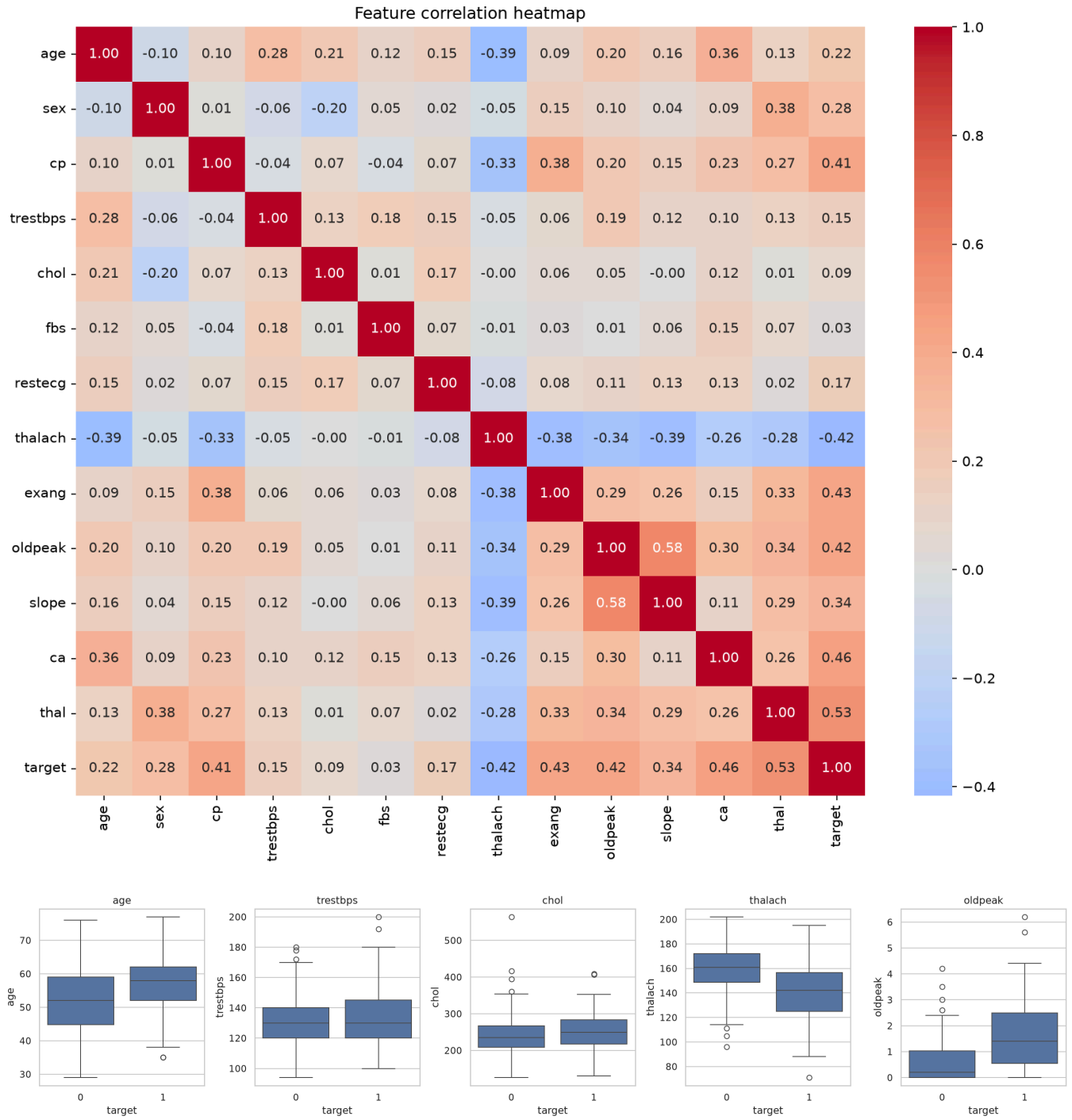
The EDA notebook (`notebooks/01_eda.ipynb`) is executed and saved with its outputs. Its main findings:

- `thal` (r = 0.53), `ca` (0.46), `exang` (0.43), `oldpeak` (0.43) and `cp` (0.41) correlate most positively with disease. `thalach` (maximum heart rate) is the only strong negative correlation (r = -0.42): patients with disease average 139 bpm against 158 bpm without.
- Chest pain type 4 (asymptomatic) has the highest disease rate, 73% of 144 patients, so `cp` is treated as a category and not as an ordered scale.
- `oldpeak` (skew 1.27) and `chol` (skew 1.14, maximum 564) are right-skewed, which supports scaling the numeric features. The numeric features are only weakly correlated with each other (largest |r| = 0.39,

`age` with `thalach`).
- `fbs` (r = 0.03) and `chol` (r = 0.09) are almost unrelated to the target on their own.
- The sample is 68% male (206 of 303), which limits how far the results generalise.


Target class balance
164
139
160
140
120
100
80
60
40
20
0
count
no disease (0)
disease (1)
target


age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target




# 4. Preprocessing and feature engineering

All preprocessing lives **inside** the scikit-learn pipeline, so it is fitted on the training folds only and cannot leak information from validation or test data.

- **Numeric features** (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`): median imputation, then `StandardScaler`.
- **Categorical codes** (`sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, `thal`): most-frequent imputation, then `OneHotEncoder(handle_unknown="ignore")`, so unseen categories at inference do not crash the API.
- **Split:** a stratified 80/20 train/test split with seed 42 (242 training and 61 test rows). The test set is never used to choose a model.

Training starts from the raw CSV, because the imputation happens in the pipeline. The cleaned copy from `clean.py` is an inspectable artefact of the data-quality step.

# 5. Model development, tuning and evaluation

Two baselines were trained: **Logistic Regression** (`class_weight="balanced"`, `max_iter=2000`) and a **Random Forest** (300 trees, `min_samples_leaf=2`, `class_weight="balanced"`). Each was evaluated with stratified 5-fold cross-validation (accuracy, precision, recall, F1, ROC-AUC) and then once on the held-out test set. The selection metric is the **mean cross-validated ROC-AUC**.

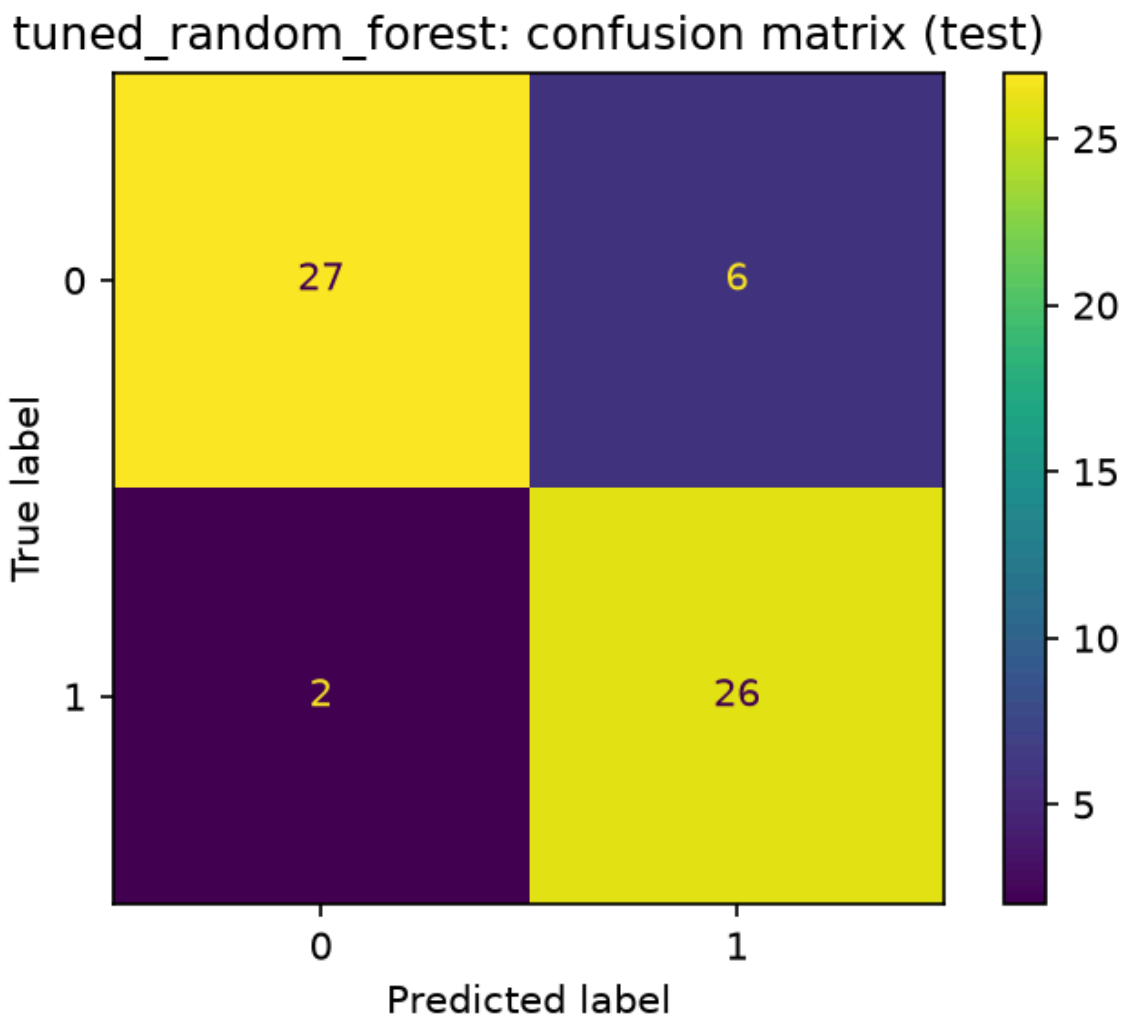
Tuning used `GridSearchCV` with the same 5-fold split and ROC-AUC scoring:

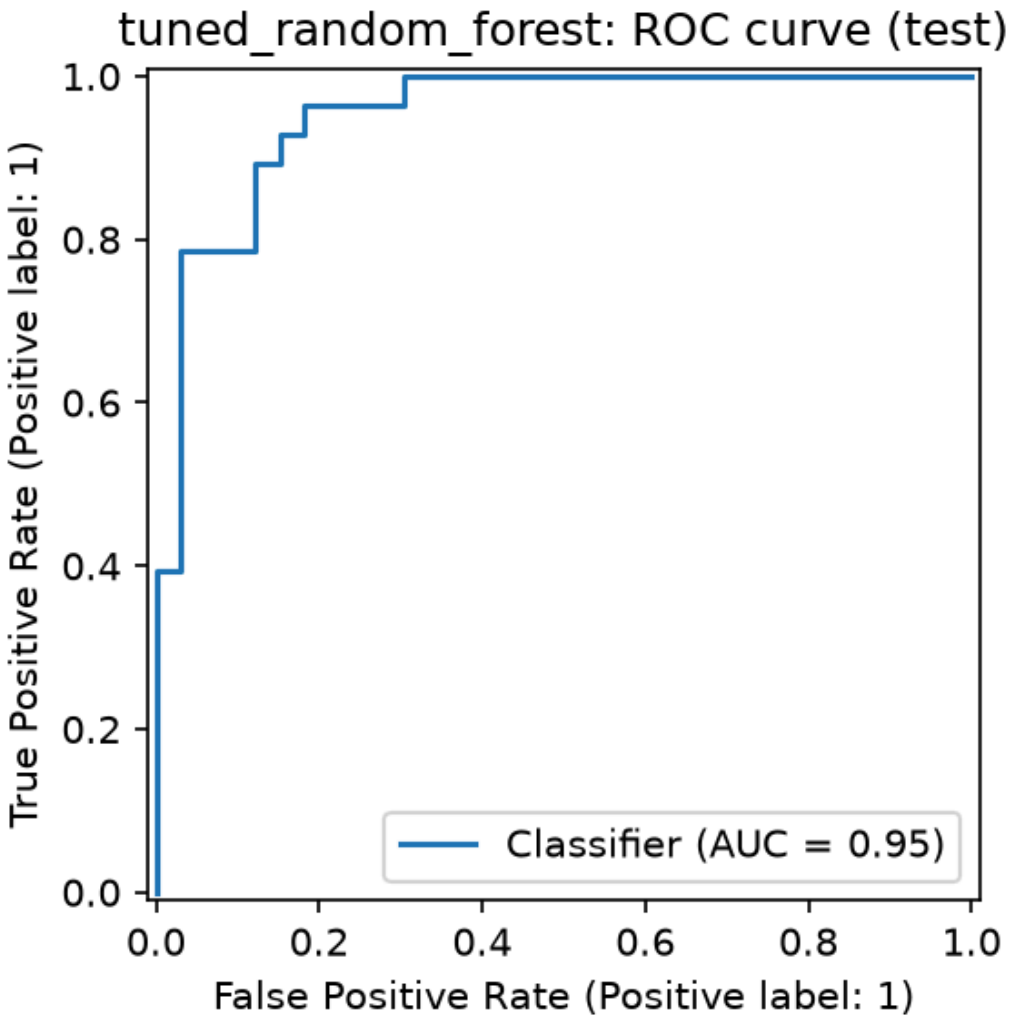
- Logistic Regression: `C` in {0.01, 0.1, 1, 10} and solver in {lbfgs, liblinear}, 8 combinations.
- Random Forest: `n_estimators` in {200, 400}, `max_depth` in {None, 6, 12}, `min_samples_leaf` in {1, 2, 4}, `max_features` in {sqrt, log2}, 36 combinations.

A tuned model replaces the baseline only if its CV ROC-AUC is higher.

| Model | CV ROC-AUC | Test Acc | Test Precision | Test Recall | Test F1 | Test ROC-AUC |
|---|---|---|---|---|---|---|
| Logistic Regression (baseline) | 0.9025 | 0.869 | 0.813 | 0.929 | 0.867 | 0.966 |
| Random Forest (baseline) | 0.8955 | 0.902 | 0.844 | 0.964 | 0.900 | 0.950 |
| Tuned Logistic Regression (C = 0.1, liblinear) | 0.9031 | 0.885 | 0.839 | 0.929 | 0.881 | 0.966 |
| **Tuned Random Forest (deployed)** | **0.9052** | 0.869 | 0.813 | 0.929 | 0.867 | 0.952 |

The tuned Random Forest uses `n_estimators` = 400, `max_depth` = 6, `min_samples_leaf` = 2 and `max_features` = log2.


tuned_random_forest: confusion matrix (test)
True label
Predicted label
0
1
0
1
27
6
2
26
25
20
15
10
5



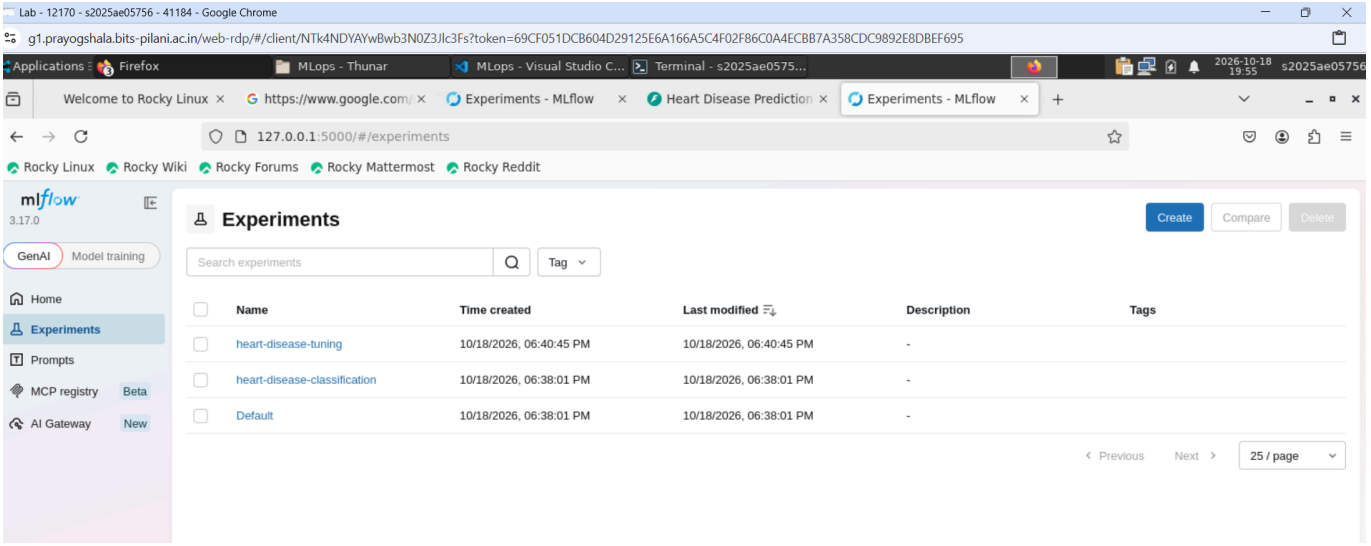

**Chosen model: the tuned Random Forest**, because it has the highest cross-validated ROC-AUC (0.9052) and it beat the baseline CV score, which is the promotion rule I set before running the experiments. I note two honest caveats. First, the gap to Logistic Regression is tiny (0.9052 against 0.9025 for the baseline and 0.9031 for the tuned version), so the two model families are effectively tied on this data. Second, the test set has only 61 rows, so one patient changes accuracy by about 1.6 points; this is why the test ranking differs from the CV ranking and why I did not select on the test set. Because the top candidates are so close, I pinned the library versions so that the result is reproducible.

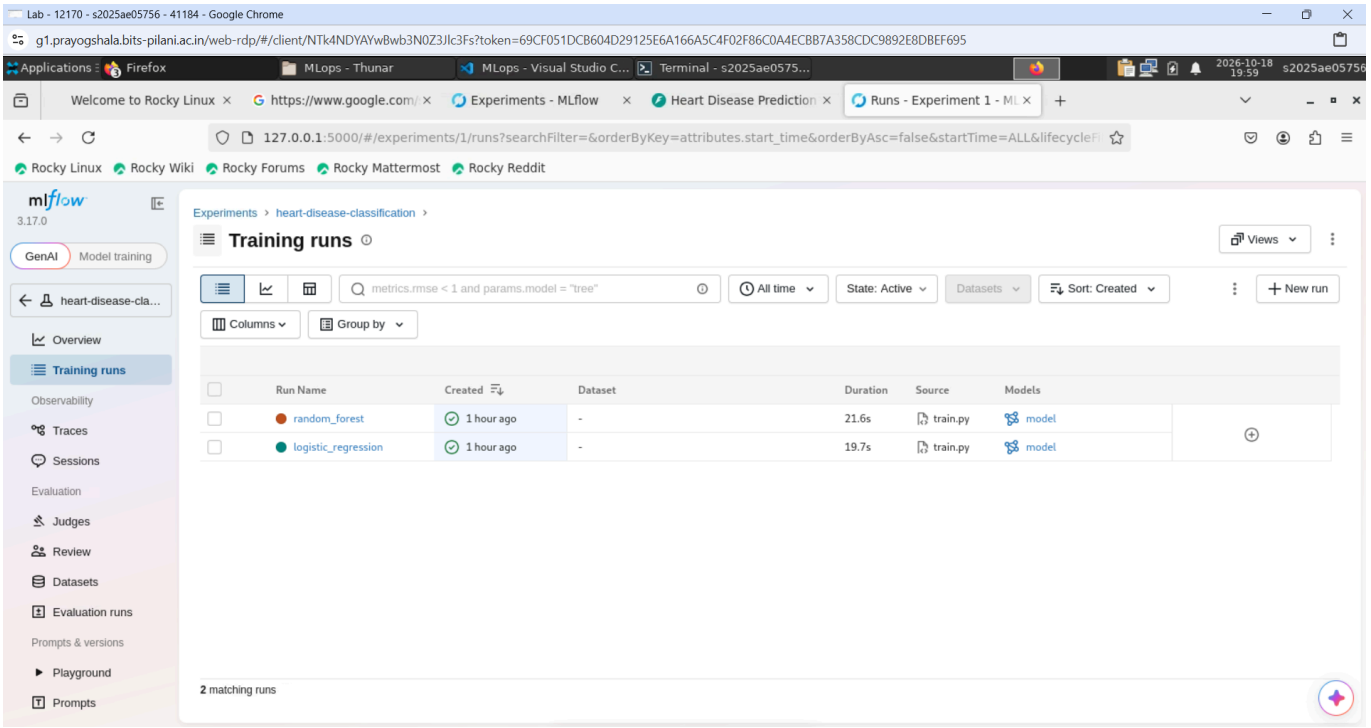
## 6. MLflow experiment tracking

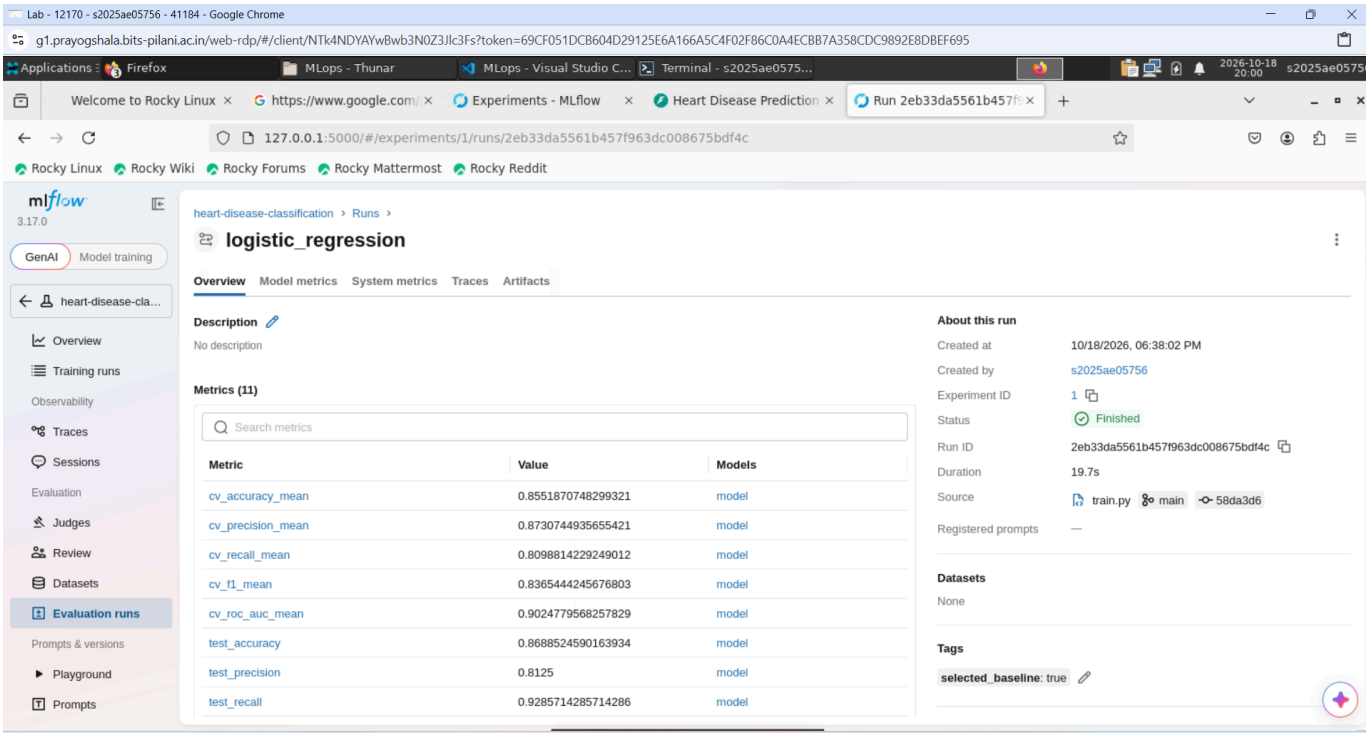
`src/train.py` and `src/tune.py` log to the same MLflow backend (`sqlite:///mlflow.db`, with artifacts under `mlruns/`), in two experiments: `heart-disease-classification` (the two baselines) and `heart-disease-tuning` (the two tuned models). Every run logs:

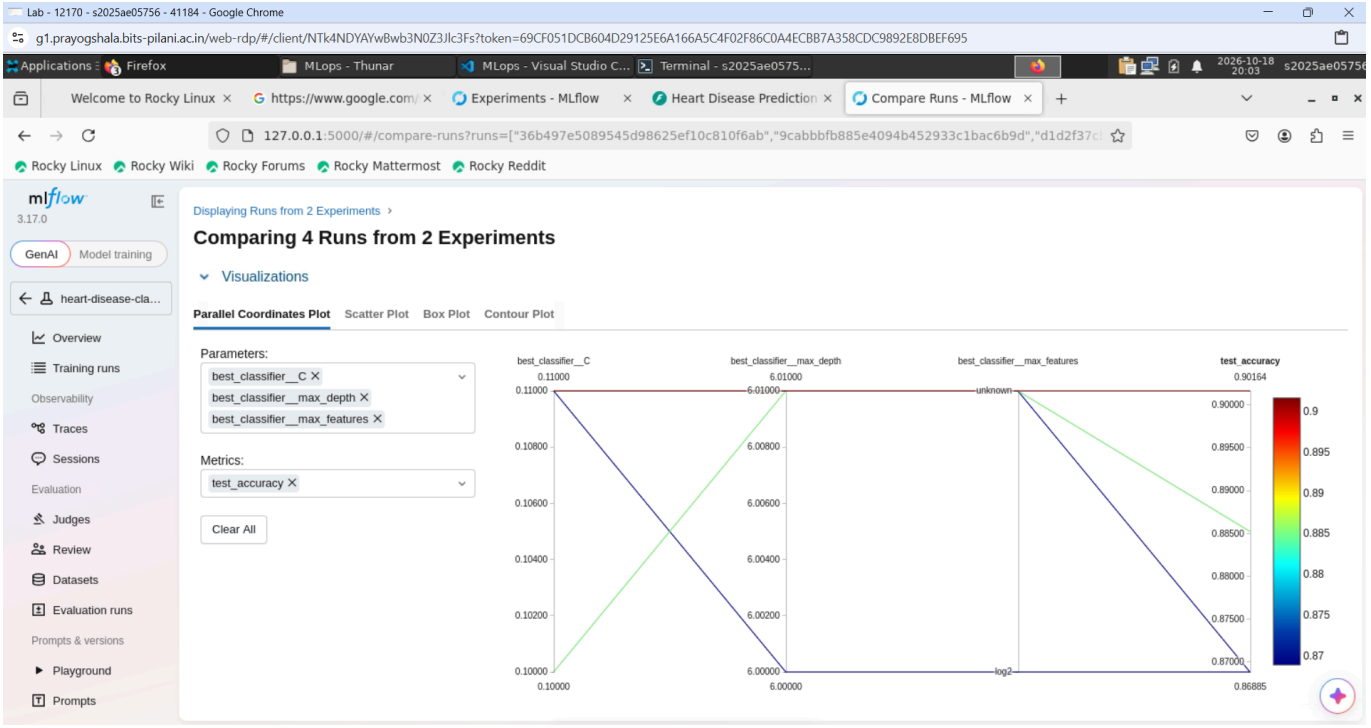
- **Parameters:** model name, test size, random state, number of CV folds and the classifier hyperparameters.
- **Metrics:** five cross-validated scores and five held-out test scores.
- **Artifacts:** confusion matrix, ROC curve, classification report, the EDA plots, `requirements.txt`, the full grid-search results as CSV (tuning runs) and the fitted pipeline.

The best baseline run is tagged `selected_baseline`, and `reports/final_model.json` records which model is deployed and where it came from.


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g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/NTk4NDYAYwBwb3N0Z3Jlc3Fs?token=69CF051DCB604D29125E6A166A5C4F02F86C0A4ECBB7A358CDC9892E8DBEF695
Welcome to Rocky Linux
Experiments - MLflow
Heart Disease Prediction
127.0.0.1:5000/#/experiments
Rocky Linux
Rocky Wiki
Rocky Forums
Rocky Mattermost
Rocky Reddit
mlflow
3.17.0
GenAI
Model training
Home
Experiments
Prompts
MCP registry
Beta
AI Gateway
New
Experiments
Create
Compare
Delete
Search experiments
Tag
Name
Time created
Last modified
Description
Tags
heart-disease-tuning
10/18/2026, 06:40:45 PM
10/18/2026, 06:40:45 PM
heart-disease-classification
10/18/2026, 06:38:01 PM
10/18/2026, 06:38:01 PM
Default
10/18/2026, 06:38:01 PM
10/18/2026, 06:38:01 PM
Previous
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g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/NTk4NDYAYwBwb3N0Z3Jlc3Fs?token=69CF051DCB604D29125E6A166A5C4F02F86C0A4ECBB7A358CDC9892E8DBEF695
Runs - Experiment 1 - ML
127.0.0.1:5000/#/experiments/1/runs?searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFi
Experiments
heart-disease-classification
Training runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
State: Active
Datasets
Sort: Created
New run
Columns
Group by
Run Name
Created
Dataset
Duration
Source
Models
random_forest
1 hour ago
21.6s
train.py
model
logistic_regression
1 hour ago
19.7s
train.py
model
2 matching runs
Overview
Training runs
Traces
Sessions
Judges
Review
Datasets
Evaluation runs
Playground
Prompts


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g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/NTk4NDYAYwBwb3N0Z3Jlc3Fs?token=69CF051DCB604D29125E6A166A5C4F02F86C0A4ECBB7A358CDC9892E8DBEF695
Run 2eb33da5561b457f
127.0.0.1:5000/#/experiments/1/runs/2eb33da5561b457f963dc008675bdf4c
heart-disease-classification
Runs
logistic_regression
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (11)
Search metrics
Metric
Value
Models
cv_accuracy_mean
0.8551870748299321
model
cv_precision_mean
0.8730744935655421
model
cv_recall_mean
0.8098814229249012
model
cv_f1_mean
0.8365444245676803
model
cv_roc_auc_mean
0.9024779568257829
model
test_accuracy
0.8688524590163934
model
test_precision
0.8125
model
test_recall
0.9285714285714286
model
About this run
Created at
10/18/2026, 06:38:02 PM
Created by
s2025ae05756
Experiment ID
1
Status
Finished
Run ID
2eb33da5561b457f963dc008675bdf4c
Duration
19.7s
Source
train.py
main
58da3d6
Registered prompts
Datasets
None
Tags
selected_baseline: true




MLflow artifacts

## 7. Packaging and reproducibility

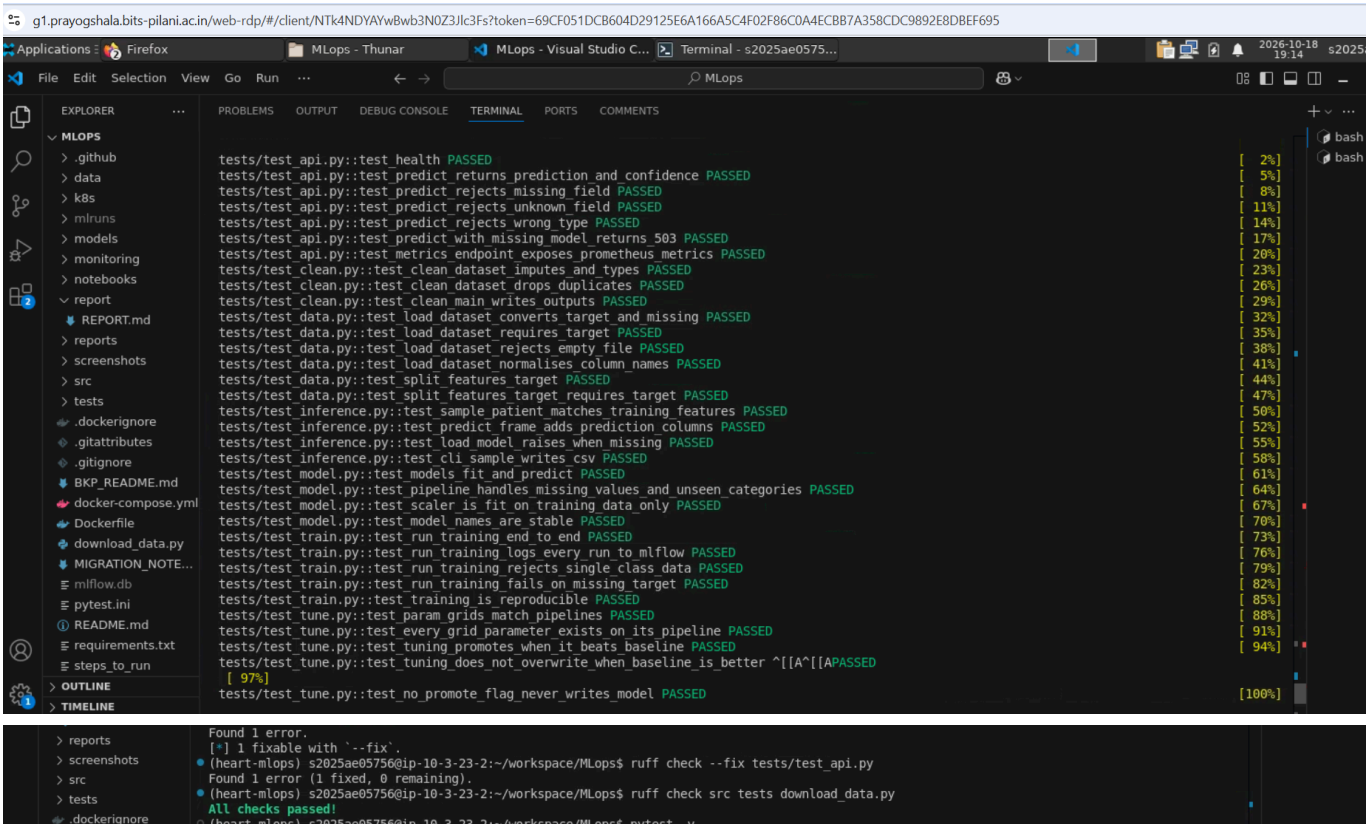
The final pipeline, including the preprocessing, is saved with joblib as `models/heart_disease_model.joblib` and is the single artefact used by the API, the Docker image and the Kubernetes deployment. A single `requirements.txt` serves the API, training, tests, notebook execution, CI and Docker. The core scientific libraries are pinned (numpy 2.4.2, pandas 2.3.3, scikit-learn 1.8.0, scipy 1.17.1, mlflow 3.17.0) and all random seeds are fixed. To reproduce from scratch:

```
pip install -r requirements.txt
python download_data.py && python -m src.clean
python -m src.train && python -m src.tune
```

Batch inference is also available from the command line: `python -m src.inference --sample` or `--input patients.csv --output predictions.csv`.

## 8. Automated tests and CI/CD

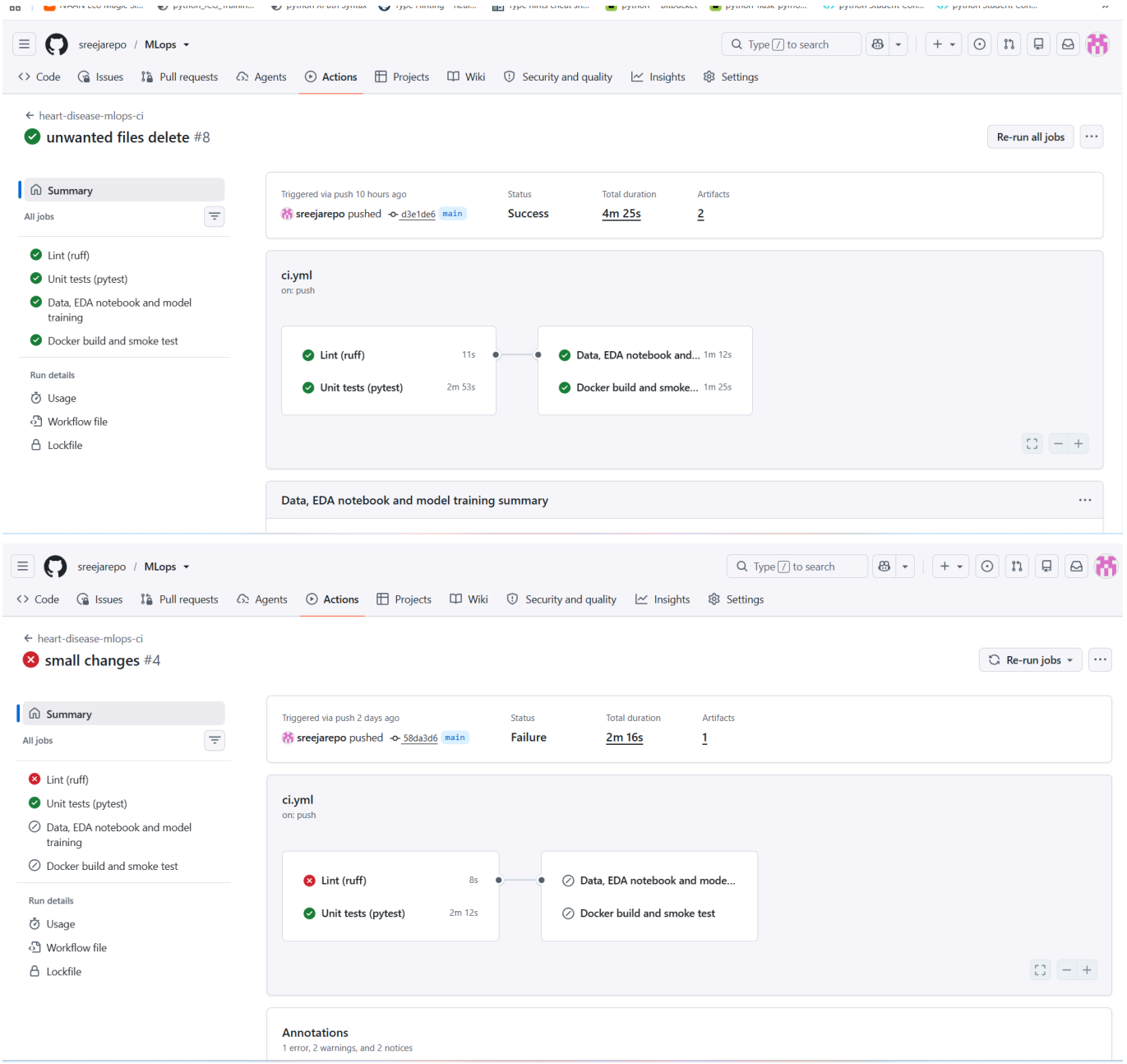
The test suite has **34 tests in 7 files**: data loading and validation (6), cleaning (3), model and preprocessing contracts (4), training and MLflow logging (5), tuning and the promotion rule (5), batch inference (4) and the API endpoints and schemas (7). The tests use small synthetic datasets and an isolated MLflow store, so they do not touch the real runs. `ruff` is used for linting and passes cleanly.




The GitHub Actions workflow `ci.yml` runs on every push and pull request with four jobs:

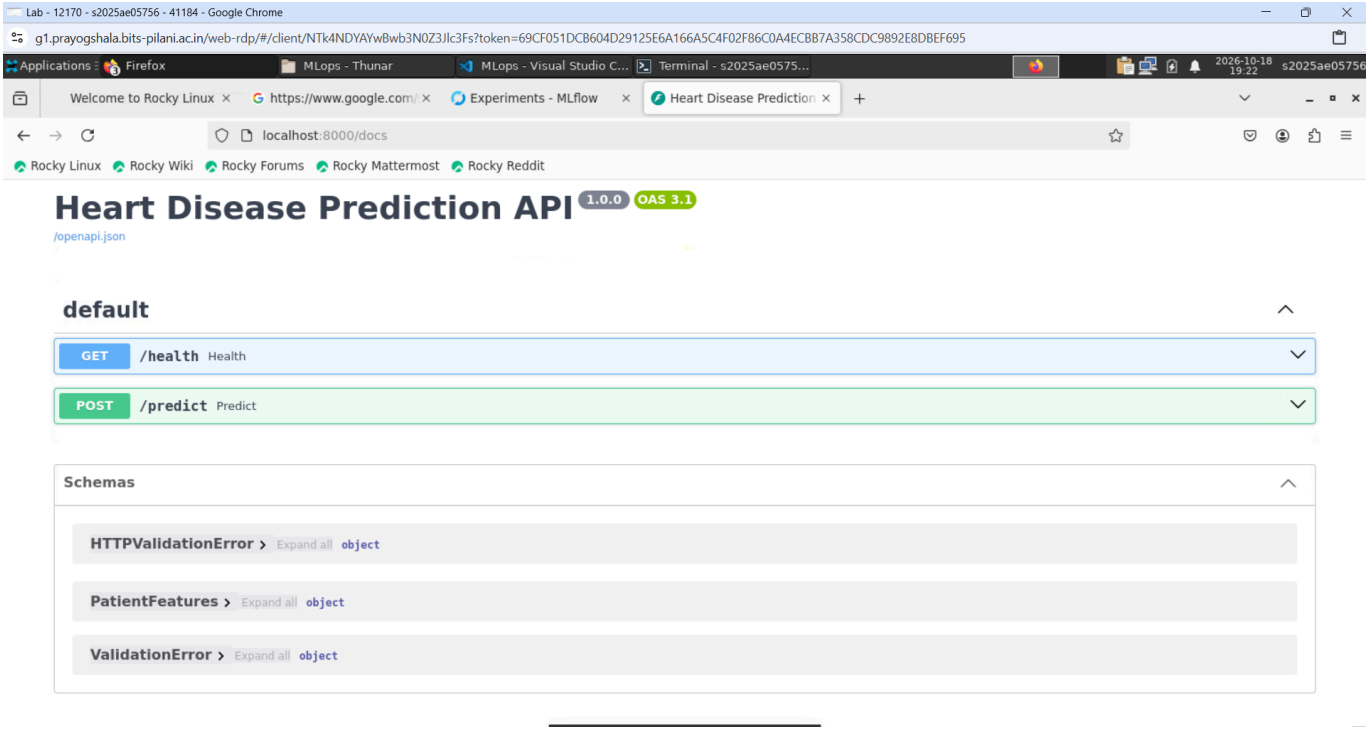
1. **lint:** `ruff check`.
2. **test:** `pytest`, with the JUnit report uploaded as an artifact.
3. **train** (after lint and test): download the dataset, build the cleaned dataset, execute the EDA notebook, train with MLflow logging, publish a metrics table to the job summary and upload the artifacts.
4. **docker** (after lint and test): build the image, start the container and call `/health`, `/predict` (the response must contain `confidence`) and `/metrics/`, then print the container logs.

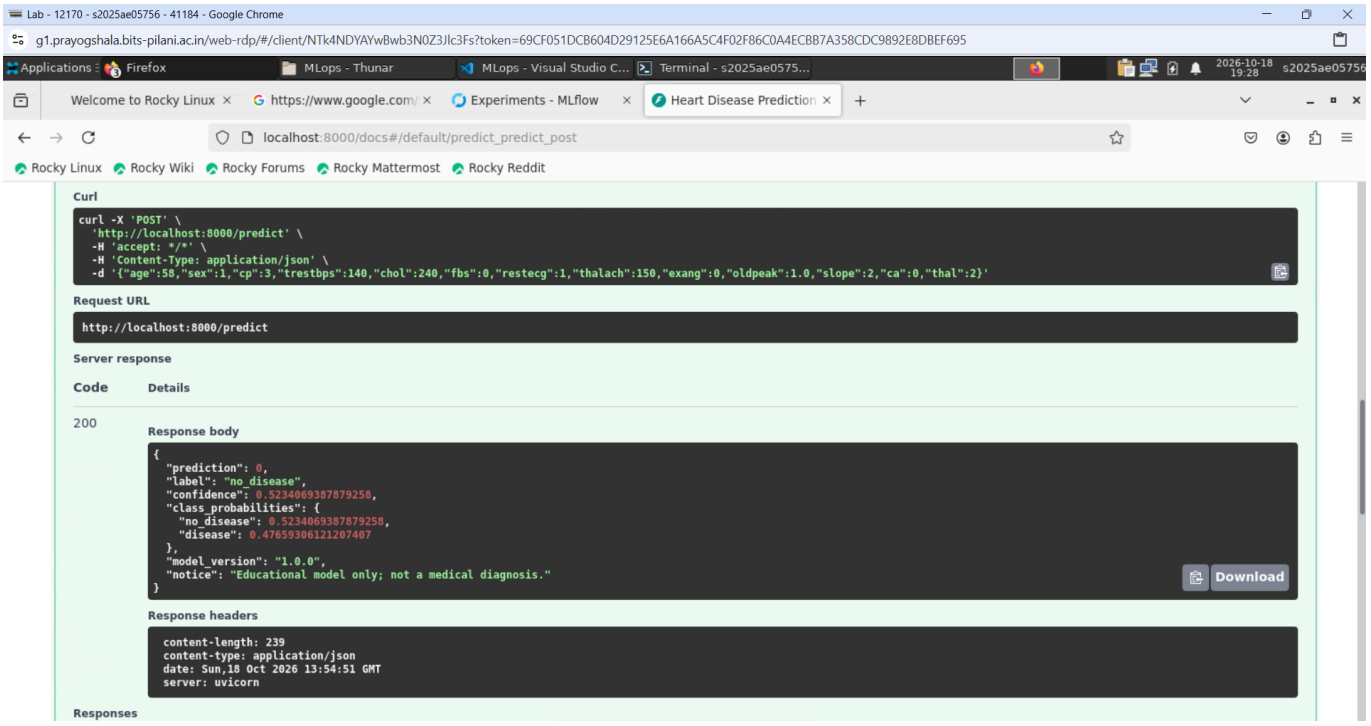
A failing step fails the workflow. A second workflow, `tune.yml`, runs the hyperparameter search on demand. To confirm that the pipeline really blocks bad changes, I also pushed a deliberate lint error on a scratch branch and saved the failed run.



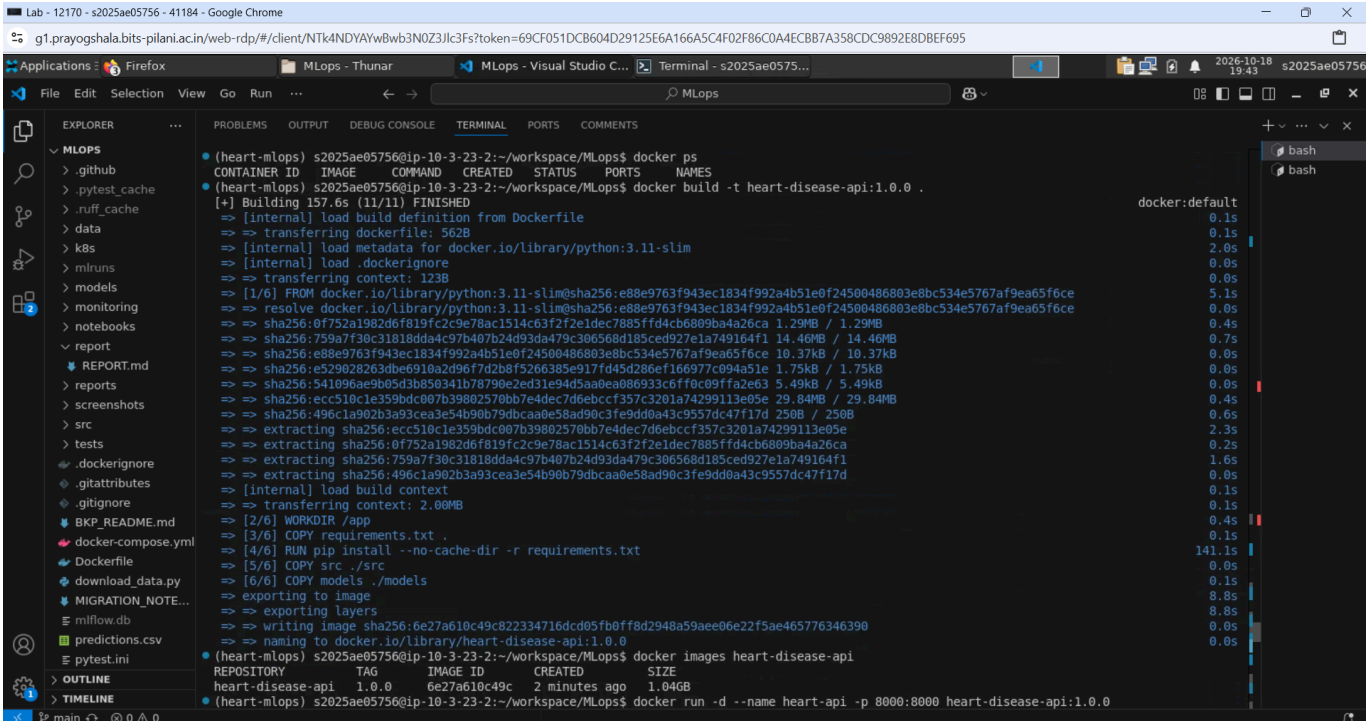

# 9. API, containerisation and Kubernetes deployment

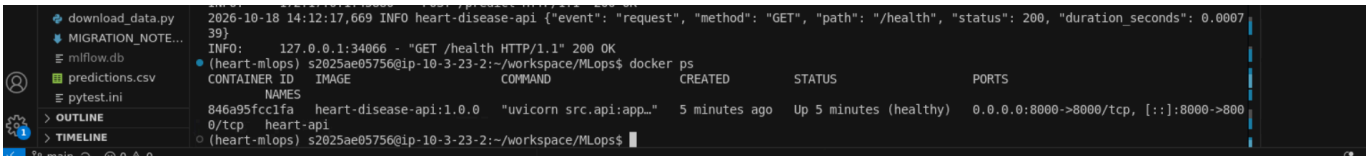
**API.** The FastAPI service (`src/api.py`) exposes `POST /predict`, `GET /health` (used by the Kubernetes probes), `GET /metrics/` (Prometheus) and the interactive `/docs`. The request body is validated against a Pydantic schema that rejects unknown fields and wrong types with HTTP 422. A response contains `prediction`, `label`, `confidence` (the probability of the predicted class), the class probabilities, the model version and a notice that it is not a medical diagnosis. Each request and each prediction is also written as a JSON log line.

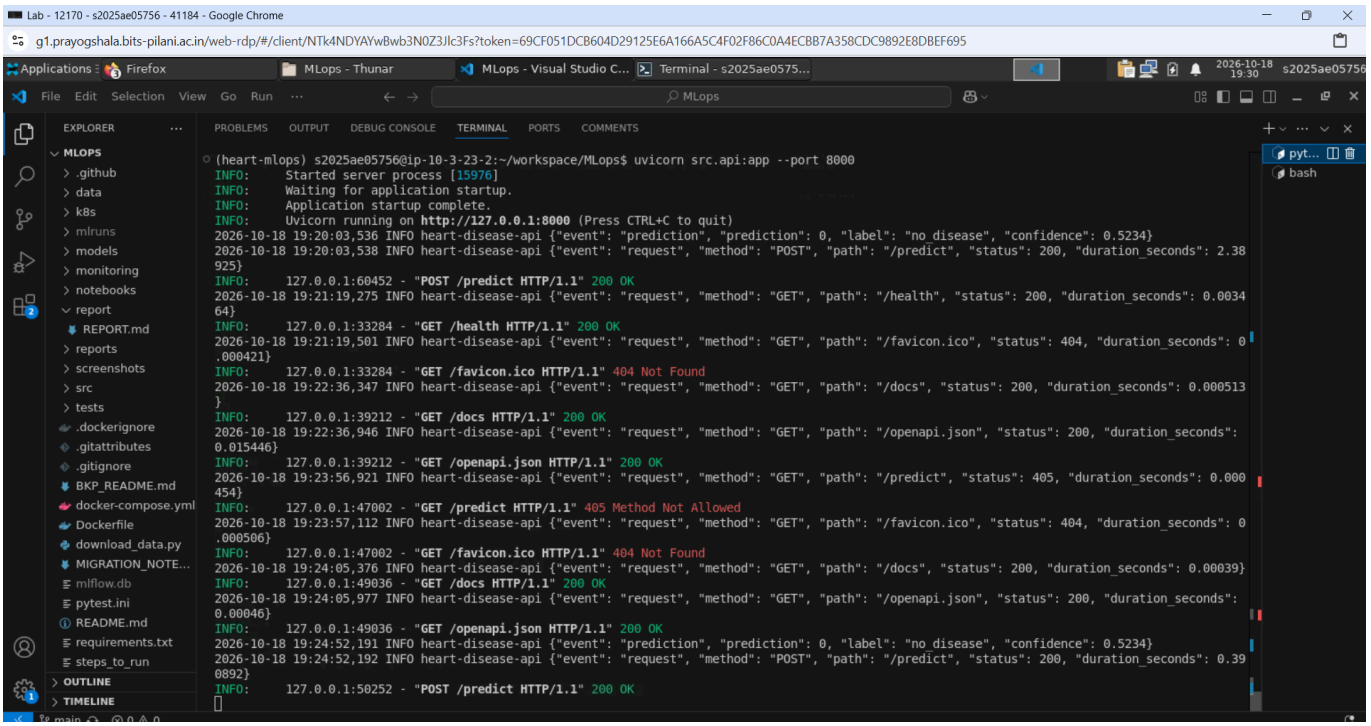


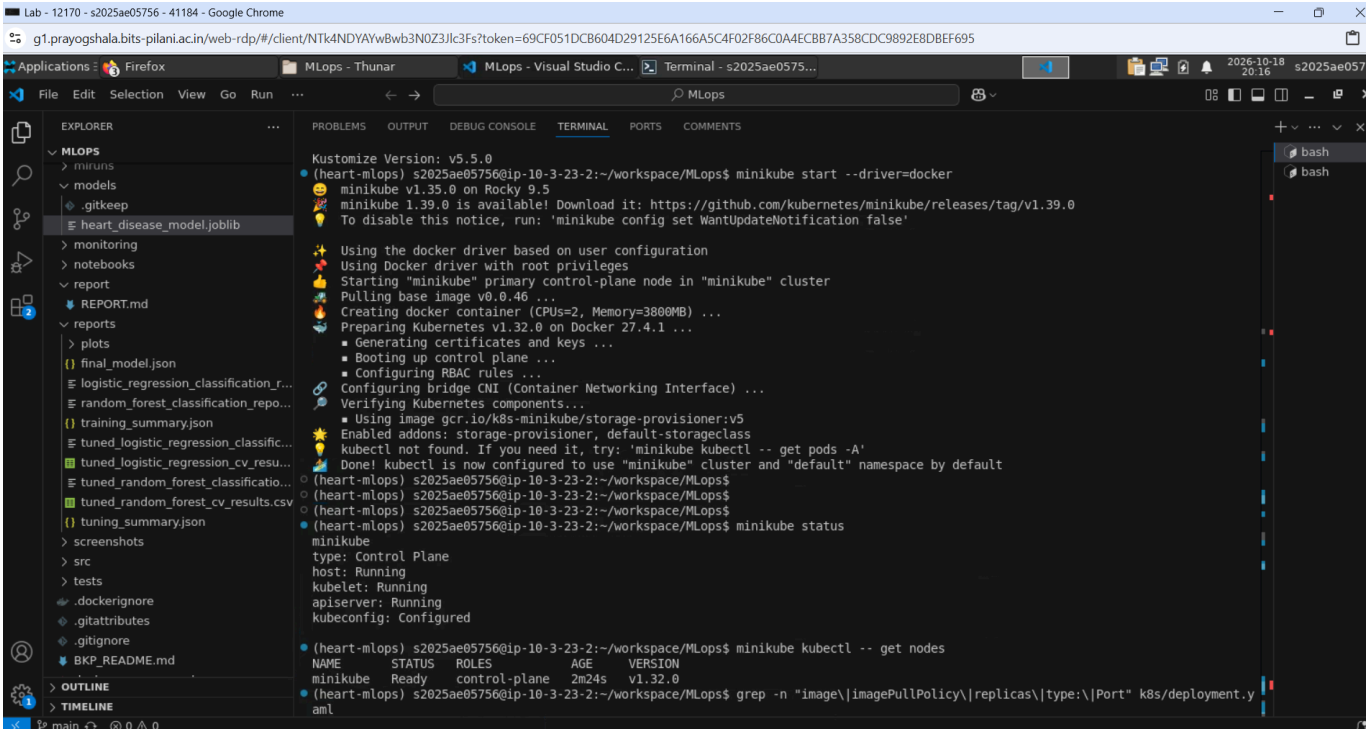

**Docker.** The image `heart-disease-api:1.0.0` is built from `python:3.11-slim`, installs `requirements.txt`, copies the source and the model, defines a `HEALTHCHECK` on `/health` and starts `uvicorn` on port 8000.

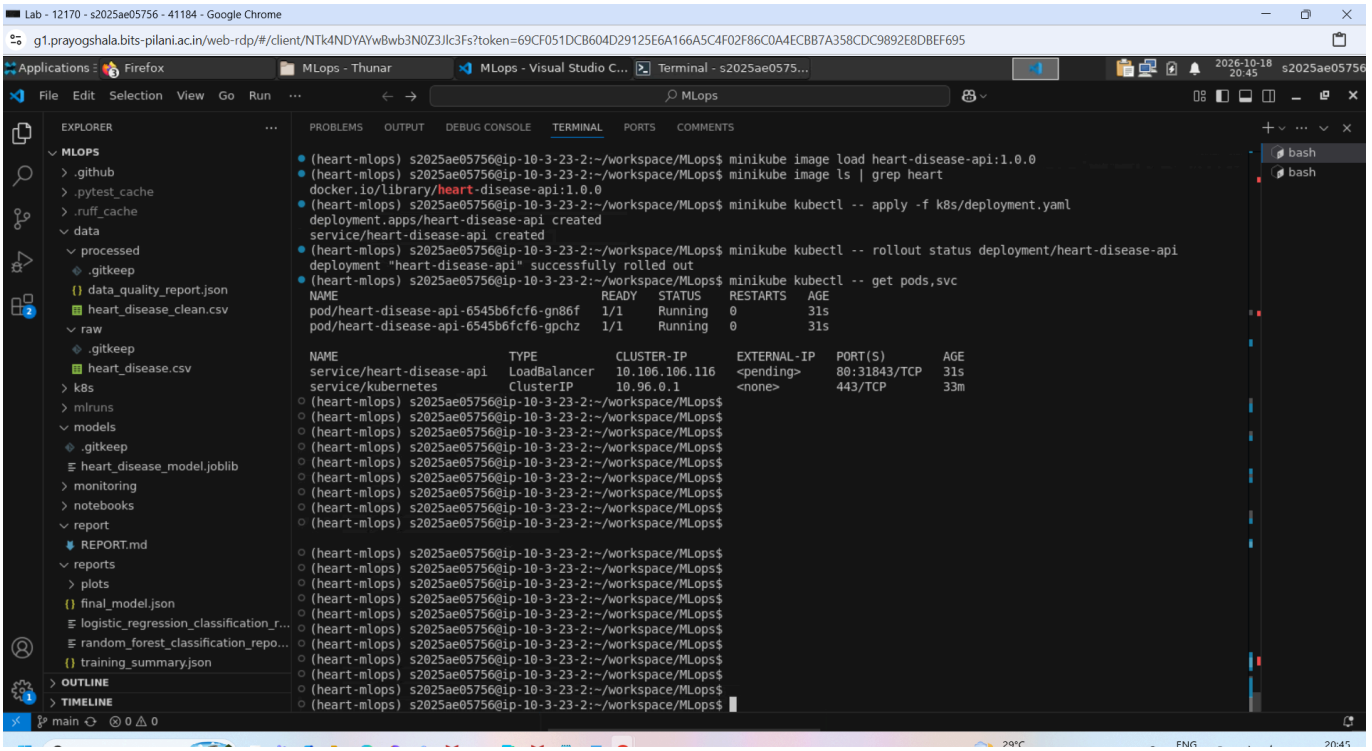


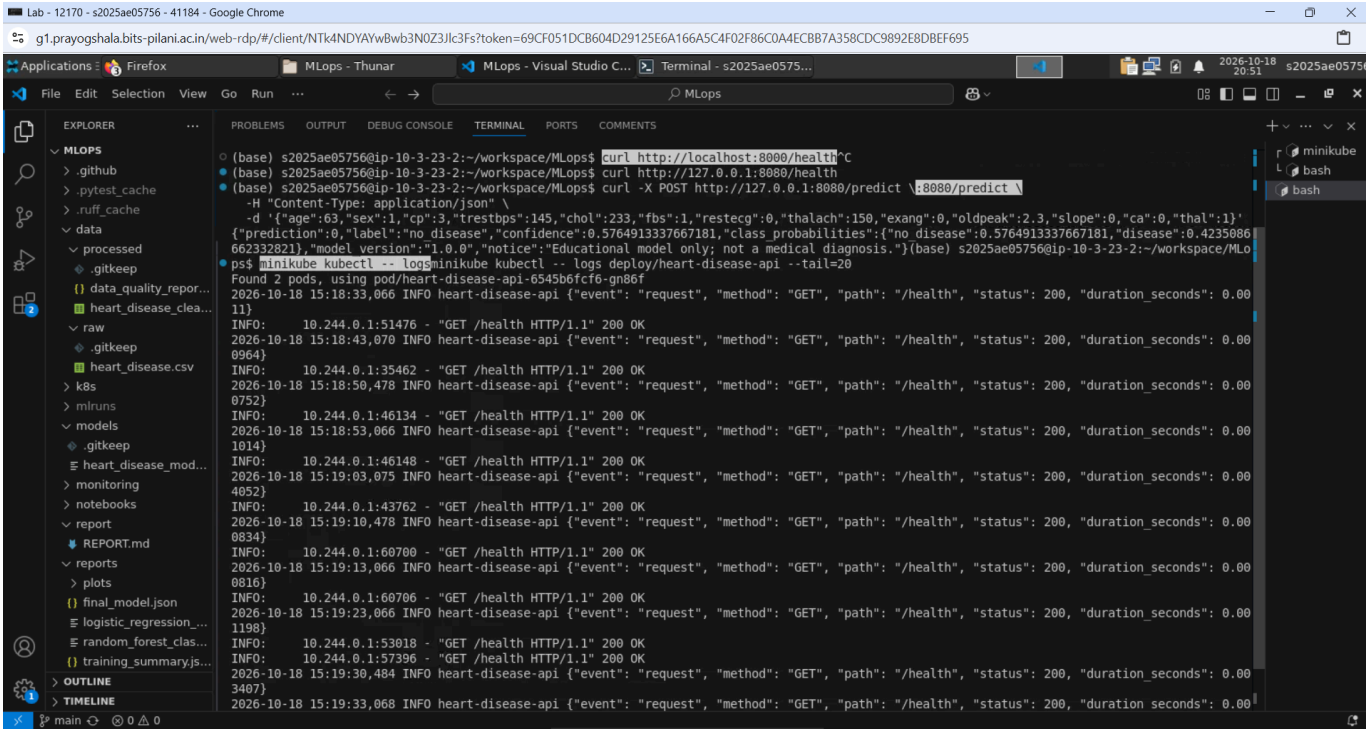




**Kubernetes (Minikube).** `k8s/deployment.yaml` defines a Deployment with **2 replicas** running the image, with readiness and liveness probes on `/health`, resource requests (100m CPU, 256Mi memory) and limits (500m CPU, 768Mi memory), and a Service of type **LoadBalancer** that maps port 80 to container port 8000. I started the cluster with `minikube start --driver=docker`, loaded the locally built image into it with `minikube image load` (the image is not in a registry, so the manifest uses `imagePullPolicy: IfNotPresent`), and applied the manifest. Both pods reached `Running 1/1`. Because a local cluster has no cloud load balancer, I reached the service with `kubectl port-forward` (an alternative is `minikube tunnel`) and called `/health` and `/predict` through it. The container logs show the same JSON log lines as the local run.


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g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/NTk4NDYAYwBwb3N0Z3Jlc3Fs?token=69CF051DCB604D29125E6A166A5C4F02F86C0A4ECBB7A358CDC9892E8DBEF695
Applications Firefox MLops - Thunar MLops - Visual Studio C... Terminal - s2025ae0575...
File Edit Selection View Go Run MLops
EXPLORER
MLOPS
models
.gitkeep
heart_disease_model.joblib
monitoring
notebooks
report
REPORT.md
reports
plots
final_model.json
logistic_regression_classification_r...
random_forest_classification_repo...
training_summary.json
tuned_logistic_regression_classific...
tuned_logistic_regression_cv_resu...
tuned_random_forest_classificatio...
tuned_random_forest_cv_results.csv
tuning_summary.json
screenshots
src
tests
.dockerignore
.gitattributes
.gitignore
BKP_README.md
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
Kustomize Version: v5.5.0
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube start --driver=docker
minikube v1.35.0 on Rocky 9.5
minikube 1.39.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.39.0
To disable this notice, run: 'minikube config set WantUpdateNotification false'
Using the docker driver based on user configuration
Using Docker driver with root privileges
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.46 ...
Creating docker container (CPUs=2, Memory=3800MB) ...
Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
Generating certificates and keys ...
Booting up control plane ...
Configuring RBAC rules ...
Configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass
kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube kubectl -- get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 2m24s v1.32.0
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ grep -n "image\|imagePullPolicy\|replicas\|type:\|Port" k8s/deployment.y
aml
bash
bash
main


Lab - 12170 - s2025ae05756 - 41184 - Google Chrome
g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/NTk4NDYAYwBwb3N0Z3Jlc3Fs?token=69CF051DCB604D29125E6A166A5C4F02F86C0A4ECBB7A358CDC9892E8DBEF695
Applications Firefox MLops - Thunar MLops - Visual Studio C... Terminal - s2025ae0575...
File Edit Selection View Go Run MLops
EXPLORER
MLOPS
.github
.pytest_cache
.ruff_cache
data
processed
.gitkeep
data_quality_report.json
heart_disease_clean.csv
raw
.gitkeep
heart_disease.csv
k8s
mlruns
models
.gitkeep
heart_disease_model.joblib
monitoring
notebooks
report
REPORT.md
reports
plots
final_model.json
logistic_regression_classification_r...
random_forest_classification_repo...
training_summary.json
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube image load heart-disease-api:1.0.0
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube image ls | grep heart
docker.io/library/heart-disease-api:1.0.0
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube kubectl -- apply -f k8s/deployment.yaml
deployment.apps/heart-disease-api created
service/heart-disease-api created
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube kubectl -- rollout status deployment/heart-disease-api
deployment "heart-disease-api" successfully rolled out
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$ minikube kubectl -- get pods,svc
NAME READY STATUS RESTARTS AGE
pod/heart-disease-api-6545b6fcf6-gn86f 1/1 Running 0 31s
pod/heart-disease-api-6545b6fcf6-gpchz 1/1 Running 0 31s
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
service/heart-disease-api LoadBalancer 10.106.106.116 <pending> 80:31843/TCP 31s
service/kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 33m
(heart-mlops) s2025ae05756@ip-10-3-23-2:~/workspace/MLops$
bash
bash
main



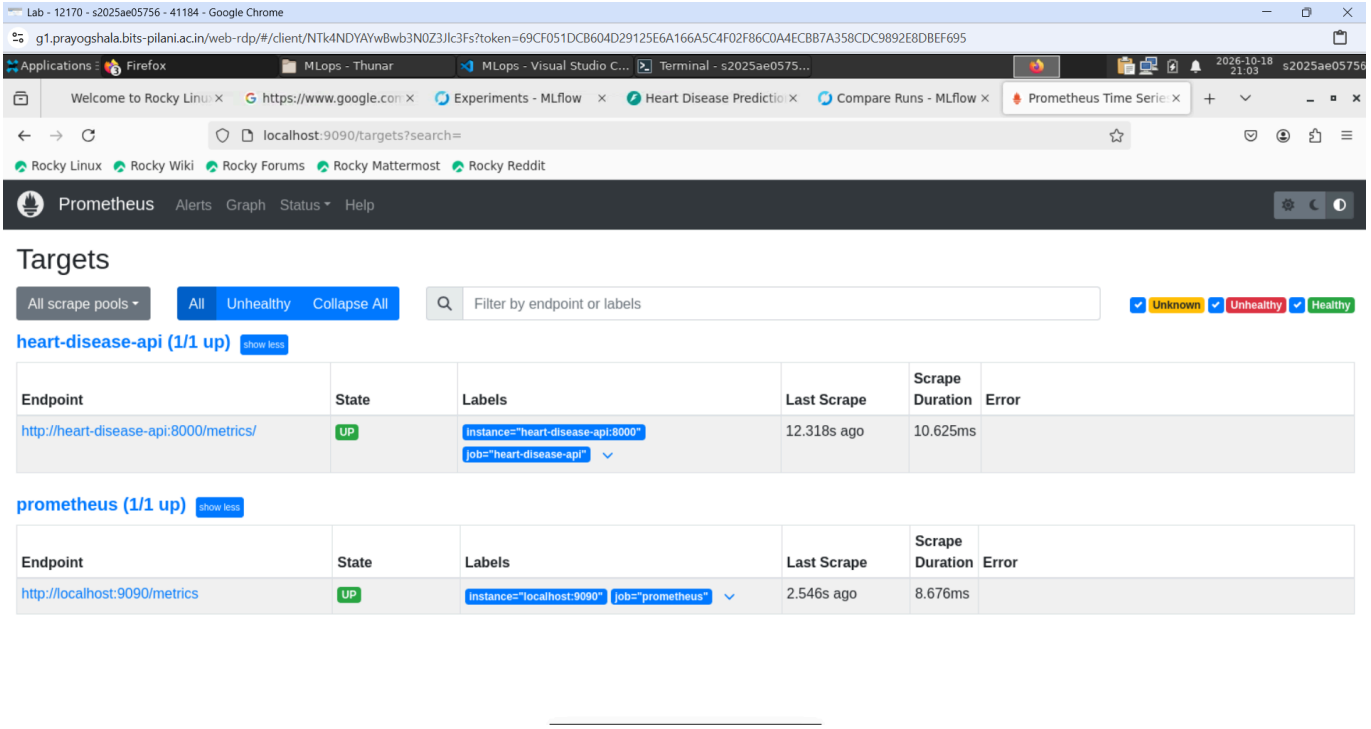

**Access instructions (local deployment):**

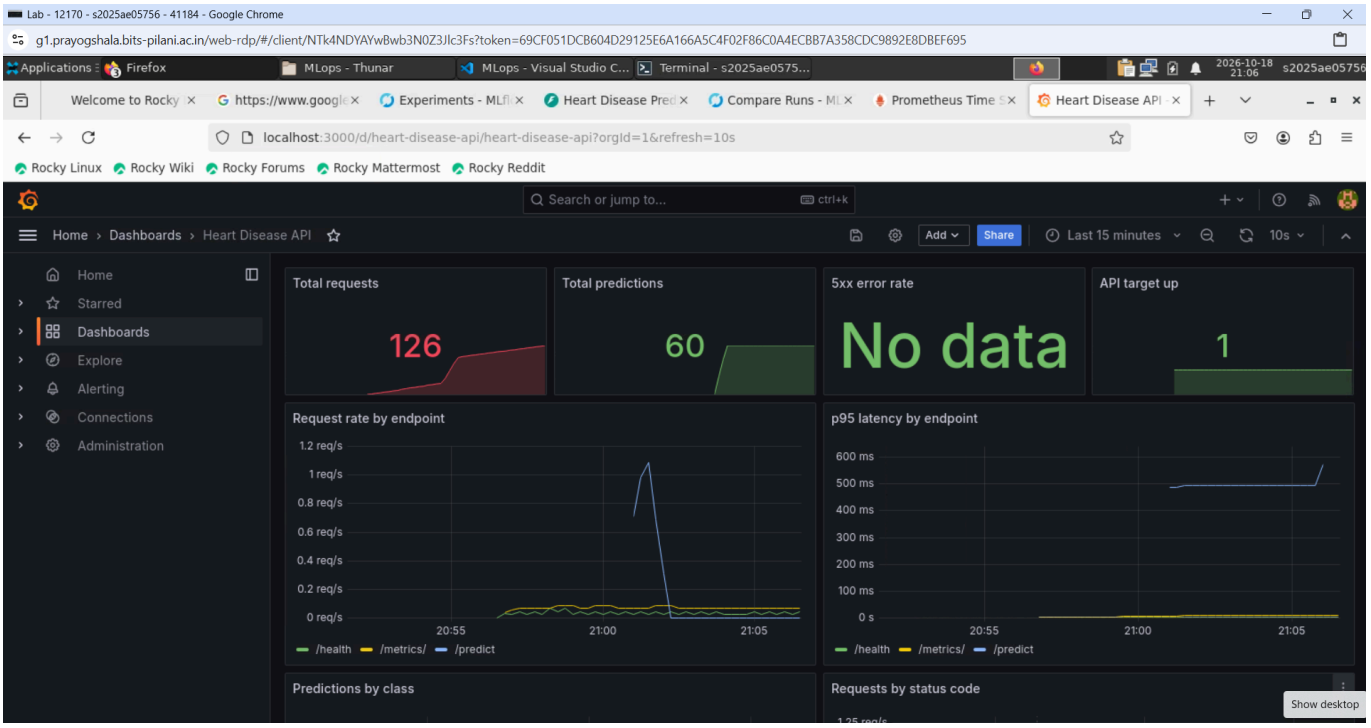
```
minikube start --driver=docker
minikube image load heart-disease-api:1.0.0
minikube kubectl -- apply -f k8s/deployment.yaml
minikube kubectl -- port-forward svc/heart-disease-api 8080:80
curl http://127.0.0.1:8080/health
```

The cluster state at the time of the run is saved in `reports/k8s_state.txt`.

## 10. Monitoring, logging, limitations and conclusion

**Monitoring.** The API exposes three custom Prometheus metrics: `api_requests_total` (labelled by method, endpoint and status), `api_request_duration_seconds` (a latency histogram per endpoint) and `predictions_total` (labelled by predicted class). `docker compose up` starts the API on port 8002, Prometheus on port 9090 (scraping `/metrics/` every 15 seconds) and Grafana on port 3000. The Grafana datasource and an 8-panel dashboard are provisioned from files: total requests, total predictions, 5xx error rate, API target up, request rate by endpoint, p95 latency by endpoint, predictions by class and requests by status code. After I sent 60 test predictions, the dashboard showed 126 total requests (the rest being health checks and metric scrapes), 60 predictions and the API target up. The 5xx panel is empty because the service returned no server errors.




**Logging.** The service writes one JSON log line per request (method, path, status and duration) and one per prediction. These are visible with `docker logs`, `docker compose logs api` or `kubectl logs`.

**Limitations.** The dataset is small (303 rows), comes from a single site (Cleveland) and is 68% male. The held-out test set has only 61 rows, so test metrics are noisy, and the model has had no external or clinical validation. The correlations found in the EDA are associations, not causal effects. The `confidence` value is the model's probability for the predicted class and is not a calibrated clinical risk. The model must not be used for diagnosis.

**Future work.** Data and model drift detection, alerting rules in Prometheus, a model registry with staged promotion, probability calibration, and validation on larger multi-site data.

**Conclusion.** The project covers the full pipeline from acquisition to monitored deployment, and each stage is automated, tested and documented. The modelling result is a near-tie between two model families, which is why selecting on cross-validation, pinning versions and tracking every run in MLflow matter more here than any single metric.

# Appendix: Architecture

